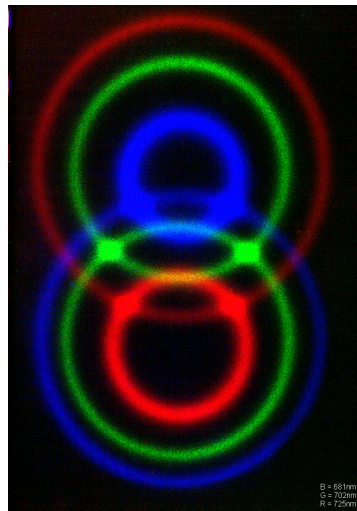


Fortgeschrittenenpraktikum 2

Entanglement and Bell's inequality



Gregor Weihs, Kaisa Laiho, Harishankar Jayakumar,
Christoph Dittel, Robert Keil

WS 2024/25

Supervisor: Johannes Loitzl
Institut für Experimentalphysik

Abstract

In this experiment you will use a nonlinear optical source of polarization entangled photon pairs to carry out a test of Bell's inequality [1]. This can answer the question whether the results of quantum measurements can be understood in a local and realistic way.

1 Introduction

Quantum mechanics predicts the possibility of nonlocal correlations between observable physical properties of distant quantum systems. This phenomenon is termed quantum entanglement. Some physicists have argued that hidden variables, which determine all properties of the systems, including measurement outcomes, but are potentially invisible to the observer, are needed to explain it. John Bell derived inequalities from the assumptions that the state of a bipartite system is fully defined via the hidden variables (“reality”) and the measurement on one system does not affect the other (“locality”) [1]. He showed that this inequality could be violated by two entangled particles, but not by particles controlled by local hidden variables. The goal of this experiment is to test Bell’s inequality using correlation measurements on polarisation entangled photon pairs generated by spontaneous parametric down-conversion (SPDC).

2 Entanglement

Erwin Schrödinger introduced the term “Entanglement” as a translation to the German word “Verschränkung” in 1935 to describe the nonlocal quantum correlations between two quantum systems that interact and then separate. In its most basic form, two qubits (i.e., two-level systems), such as the spins of electrons or nuclei or the polarization of photons, are entangled. Here we will focus only on the polarization of photons.

Let’s consider the quantum state of polarization of photons A and B. The orthogonal bases are

$$|H\rangle_{A(B)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}_{A(B)}, |V\rangle_{A(B)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}_{A(B)}$$

for photons A and B, respectively. H and V refer to horizontal (0°) and vertical polarizations (90°) of the photons with respect to a suitable frame of reference (e.g. the lab table). An example for an entangled state of the two photons is given by

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}(|H\rangle_A |H\rangle_B + |V\rangle_A |V\rangle_B)$$

If we measure the polarization of photon A and if the outcome is H , then the state of the system will collapse to $|H\rangle_A |H\rangle_B$. Any subsequent measurement in the same basis for photon B will always have H as the outcome, such that the two measurement results are not independent from another, but correlated. Formally speaking, an entangled state such as $|\phi^+\rangle$ cannot be decomposed into a tensor product of the state of photon A ($|\psi\rangle_A$) and the state of photon B ($|\phi\rangle_B$), a so-called product state ($|\psi\rangle_A \otimes |\phi\rangle_B$), which would produce independent measurement results for its sub-systems:

$$|H\rangle_A |H\rangle_B + |V\rangle_A |V\rangle_B = |H\rangle_A \otimes |H\rangle_B + |V\rangle_A \otimes |V\rangle_B \neq |\psi\rangle_A \otimes |\phi\rangle_B \forall \{|\psi\rangle_A, |\phi\rangle_B\}.$$

This gives rise to the following definition: All states of a composite system that cannot be represented as product states are entangled states.

There is a variety of measures, how much entangled a state is. The following four states,

known as the Bell states, are maximally entangled by all these measures. In the context of photon polarization, they have the following form (one being the state $|\phi^+\rangle$ from above):

$$\begin{aligned} |\psi^+\rangle &= \frac{1}{\sqrt{2}}(|HV\rangle + |VH\rangle) & |\psi^-\rangle &= \frac{1}{\sqrt{2}}(|HV\rangle - |VH\rangle) \\ |\phi^+\rangle &= \frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle) & |\phi^-\rangle &= \frac{1}{\sqrt{2}}(|HH\rangle - |VV\rangle), \end{aligned}$$

where we omitted the indices A and B for brevity. Additionally, we can express the Bell states in the D (iagonal)/ A (ntidiagonal) basis, which are linear polarizations rotated by $\pm 45^\circ$:

$$|D\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |A\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

The Bell states in this case are as follows (prove this as **exercise 1**):

$$\begin{aligned} |\psi^+\rangle &= \frac{1}{\sqrt{2}}[|DD\rangle - |AA\rangle] & |\psi^-\rangle &= \frac{1}{\sqrt{2}}[|AD\rangle - |DA\rangle] \\ |\phi^+\rangle &= \frac{1}{\sqrt{2}}[|DD\rangle + |AA\rangle] & |\phi^-\rangle &= \frac{1}{\sqrt{2}}[|DA\rangle + |AD\rangle]. \end{aligned}$$

For this basis change the $|\psi^-\rangle$ and the $|\phi^+\rangle$ state keep the same form. In general, if we change the basis for both subsystems the same way, the $|\psi^-\rangle$ or singlet state will always keep its form, whereas the other (triplet) states can change into each others' forms. In any case, no matter how we change bases, even for arbitrary choices in A and B the states always remain entangled.

Let us assume we have a system prepared in the state $|\psi^-\rangle$. If we now measure the polarization of one photon, say A, by setting our polarizer to 0° , we will find that half of the photons pass the polarizer and half are reflected (or absorbed). The same will be true for any other angle setting of the polarizer! However, if we also measure polarization of the second photon and look for simultaneous detection the story gets more complicated. If the two polarizers are parallel we will never find the two photons of a pair transmitted. It will always be one transmitted and one reflected. If, on the other hand the two polarizers are oriented orthogonally we find that the two photons of a pair will always be both transmitted or both reflected. This is a strong anticorrelation in the polarisation of the pair, which appears no matter how far apart the two measurements are taken (therefore being a nonlocal correlation).

3 Bell's inequality

In 1964 J. S. Bell derived an inequality for such correlation measurements, which would need to be satisfied by a local realistic theory based on hidden variables. Entangled quantum states, however, could violate this inequality. There are many variants of Bell's inequality; In 1969 Clauser, Horne, Shimony, and Holt (CHSH) derived a version [2] that is well suitable for experiments. It is given by

$$|E(\alpha, \beta) - E(\alpha, \beta') + E(\alpha', \beta) + E(\alpha', \beta')| \leq 2, \quad (1)$$

where $E(\alpha, \beta) = \langle ab \rangle$ is the expectation value of polarization correlation measurements made on a two-photon system with the polarization of photon A(B) measured at the angles $\alpha(\beta)$ and

the measurement results $a(b) = \pm 1$ recording whether the respective photon was transmitted and, thus, polarised in parallel to the chosen angle (+1) or not transmitted, which means perpendicular polarisation (-1). A very good exposition of this inequality was given by John Bell himself in Ref. [3] (in the context of spin-1/2 particles and a Stern-Gerlach experiment).

Here, we only provide a poor man's derivation: Suppose two observers, Alice and Bob, each receive one photon from a two-photon system. Alice can measure the polarization of her photon at the angles α and α' , while Bob can measure his photon at β and β' . For each choice of angles, both measurement devices determine whether their photons were polarised parallel (+1) or perpendicular (-1) to these angles. For example, Alice measuring a photon at $\alpha = 0^\circ$ (which we define here as H) can either get the result H , which is parallel to α (yielding $a = 1$) or V , which is perpendicular ($\alpha = 90^\circ$, $a = -1$). Let the variables a, a', b and b' represent the measurement results for the angles α, α', β and β' . Now consider the expression

$$ab - ab' + a'b + a'b' = a(b - b') + a'(b + b'). \quad (2)$$

As $b, b' = \pm 1$, either $a(b - b') = 0$ or $a'(b + b') = 0$, which means that eq. (2) must equal ± 2 . Now, if several photon pairs are tested, each pair yields ± 2 for eq. (2) in a random manner. This implies for the expectation values precisely the CHSH inequality (1). A more rigorous theoretical analysis (via integrating probabilities over the fictitious hidden variables) leads to an even more restrictive inequality

$$S = |E(\alpha, \beta) - E(\alpha, \beta')| + |E(\alpha', \beta) + E(\alpha', \beta')| \leq 2, \quad (3)$$

which contains (1), due to the triangle inequality. S is called the Bell parameter.

The CHSH inequalities can be violated, for instance, if the two photons are in any of the Bell states. In particular, if the photons are in the entangled $|\psi^-\rangle$ state, the expectation value according to quantum mechanics is given by

$$E(\alpha, \beta) = -\cos[2(\alpha - \beta)]. \quad (4)$$

Exercise 2: Show that in this case the Bell parameter S reaches $2\sqrt{2}$ for the angles $\alpha = 0^\circ + \phi$, $\beta = 22.5^\circ + \phi$, $\alpha' = 45^\circ + \phi$ and $\beta' = 67.5^\circ + \phi$ for any constant offset ϕ .

In an actual experiment “no transmission” is clearly not a good indicator for perpendicular polarisation of a photon with respect to the polariser, simply because the photon could have been absorbed elsewhere in the setup (or not created or not detected). Therefore, one instead measures the number of detected photons in a given time interval for a polariser rotated by 90° and compares it with the number of detected events for the original polariser setting. Then the expectation values $E(\alpha, \beta)$ can be calculated from the number of pair-detection events (coincidences) for each set of polarizer angles as follows:

$$E(\alpha, \beta) = \frac{1}{N}(C_{++}(\alpha, \beta) + C_{--}(\alpha, \beta) - C_{+-}(\alpha, \beta) - C_{-+}(\alpha, \beta)). \quad (5)$$

where $C_{++}(\alpha, \beta)$ represents the number of events when the polarizer A is set to α and polarizer B is set to β , C_{+-} represents the number of events when the polarizer A is set to α and polarizer B is set to $\beta + 90^\circ$, etc.. The normalisation term N is the total number of events given by

$$N = C_{++}(\alpha, \beta) + C_{--}(\alpha, \beta) + C_{+-}(\alpha, \beta) + C_{-+}(\alpha, \beta)$$

In the above argumentation we have made implicit assumptions which need some discussion. We will detect many photons in the experiment, but for most of them we will fail to detect its partner photon. Therefore, we have to assume that the hidden variables will not influence the detection efficiency and thus that the detected coincidences are a “fair sample” of all the emitted photon pairs. Another assumption is that no communication occurs between the two branches of the experiment. That is, we assume that no information on polariser settings and measurement results on Alice’s side can influence the measurement outcomes on Bob’s (via any potentially unknown process). To close this “locality loophole”, a space-like separation of the analysing apparatuses is required (i.e. detection and polariser setting on one side take place outside the light-cone of the setting and detection at the other side). Only recently three experiments [4, 5, 6] managed to achieve high enough detection efficiency together with space-like separation to be able to violate Bell’s inequality without either of these two assumptions.

4 Spontaneous parametric down-conversion

There are many sources of entangled photons, such as quantum dots, spontaneous parametric down conversion (SPDC), single trapped atoms and single trapped ions, and even DNA structures. Here, we use SPDC in a nonlinear optical crystal to produce entangled photon pairs and test the Bell inequality.

SPDC is a nonlinear optical process which relies on the second-order optical nonlinearity of certain media (requiring non-inversion-symmetric crystals); for example, KDP, LiIO₃, KNbO₃, LiNbO₃ and BBO. In this experiment, a violet (404 nm) pump laser down-converts into two infra-red fields, known as the signal and idler, inside a BBO (β - BaB₂O₄) crystal. The process is a distinctly quantum mechanical phenomenon, just like the spontaneous emission of an excited atom, and it always produces signal and idler photons in pairs. Two conservation laws have to be obeyed. The first law is the conservation of energy, which requires the frequency of the pump photon to equal the sum of the frequencies of the two generated photons.

$$\hbar\omega_{\text{pump}} = \hbar\omega_{\text{signal}} + \hbar\omega_{\text{idler}}$$

Furthermore, the photons are emitted diametrically opposite to one another around the pump direction, which is a consequence of the conservation of momentum. It requires the wavevector of the pump photon to equal the sum of the wavevectors of the two generated photons.

$$\hbar\mathbf{k}_{\text{pump}} = \hbar\mathbf{k}_{\text{signal}} + \hbar\mathbf{k}_{\text{idler}} \quad (6)$$

In general it is difficult to satisfy the momentum conservation, because all media are dispersive and the wavenumber is proportional to the frequency-dependent refractive index $n(\omega)$, i.e. $k(\omega) = \omega n(\omega)/c$. If the momentum conservation is not fulfilled, signal (and idler) photons created in different locations of the crystal acquire a different phase and interfere destructively with each other. The photons are produced in phase (one also speaks of “phase-matching”) only if (6) is satisfied. In our setup, we exploit the birefringence of the BBO crystal to achieve phase-matching. Specifically, the signal photon carries extraordinary polarization (here V), while the idler photon is ordinarily polarized (here H). This is called type-II phase-matching. As a consequence, signal and idler photons are emitted on two distinct cones, as can be seen

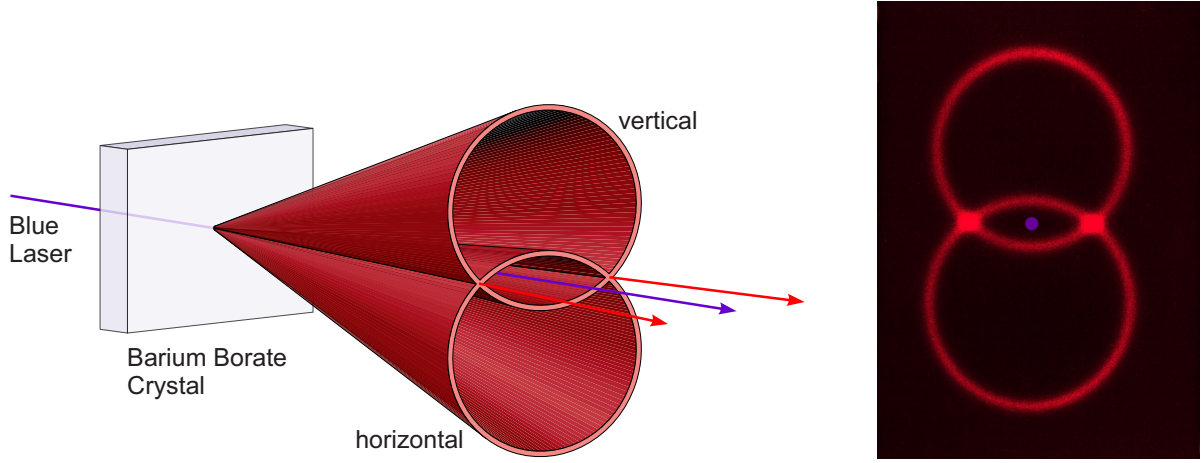


Figure 1: Left: Type-II SPDC produces photon pairs, which emerge on two cones where the vertically polarized photon is on the upper cone and the horizontally polarized photon is on the lower cone. Polarization entangled photons are observed at the two intersection lines of the cones. Right: False-color photograph of the down-conversion light taken with a bandpass filter at the degenerate wavelength.

in Figure 1.

Polarization entangled photons can be obtained by spatial filtering of the photons emerging on the intersection lines of the two cones and spectral filtering around the degeneracy frequency $\omega_{\text{signal}} = \omega_{\text{idler}} = \omega_{\text{pump}}/2$. Momentum conservation guarantees that one photon will emerge at each intersection, but it will be impossible to distinguish which cone either photon came from. However, the two photons must carry opposite polarizations.

Exercise 3: What kind of two-photon state do you expect at the intersection of the cones and why?

5 Experimental setup

Figure 2 shows the schematic setup of the entangled photon pair source. The pump laser is focused into the BBO crystal, where photon pairs are produced via SPDC. The intersection directions of the cones are separated by two small prism mirrors and then directed via mirrors (not shown) and further lenses into optical fibers to eventually detect the two photons of a pair. Due to the birefringence of the BBO crystal the propagation directions and velocities of horizontally and vertically polarized light are slightly different as they travel through the crystal. This leads to a transverse walk-off (spatial displacement) and longitudinal walk-off (temporal displacement) of the horizontally polarized photons from the vertically polarized photons. Thus the photons can be distinguished by their position and relative timing and, thus, do not contribute to the entanglement. To preserve the polarization entanglement, we will have to make sure that there is no information available that can be used to determine the polarization state of the photons. So, it is necessary to compensate the walk-off before the measurements. While this cannot be done completely, the distinguishability of the photons

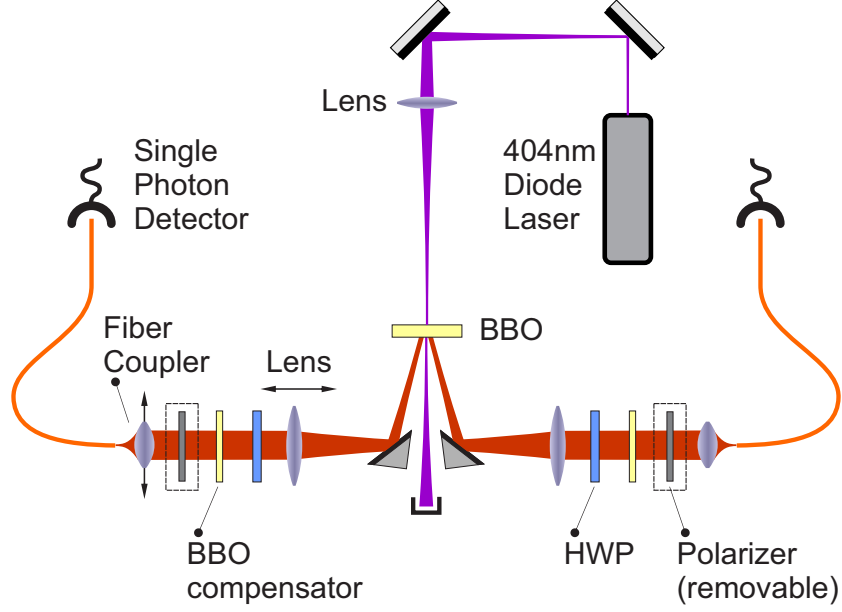


Figure 2: Schematic of the experimental setup. A blue laser pumps the BBO crystal to create photon pairs. The intersection lines of the H and V cones deviate from the pump beam by about 3° . Small prism mirrors separate the down-conversion beams, which pass a collimating lens, a half-wave-plate (HWP), the compensating BBO crystal and a removable polarizer, before getting coupled into a single-mode optical fiber for spatial selection and eventually detection. The actual setup includes color glass filters and bandpass filters for spectral selection as well as another pair of mirrors on the single photon paths (not shown here).

can be much reduced by the use of compensator BBO crystals. First, the down-converted photons have their polarizations rotated by 90° using a halfwave plate so that H and V get exchanged. Each photon is then passed through a compensator BBO crystal which is half as thick as the original crystal that performs down-conversion. In this fashion, the walk-off effects are reversed by half, which consequently erases all temporal distinguishability and reduces the effects of the transverse walk-off. The temporal compensation is illustrated in Figure 3.

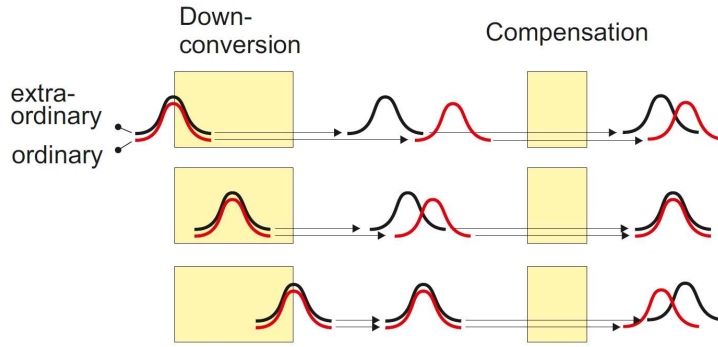


Figure 3: Walk-off compensation by reversing half of the relative time delay between ordinary and extraordinary photons of a pair. In this way the relative time delay reveals no information about the polarization state of a photon.

In the setup there are additional spectral filters and polarizers. The filters remove stray pump light and ensure that only photons of (nearly) the same wavelength are coupled into the fibers (the bandpass filters have 3 nm bandwidth). The polarizers can be inserted to perform the correlation measurements. The fibers are finally connected to single photon detectors (APD - avalanche photodiode), which trigger a macroscopic current when a photon impinges on them. The APDs are connected to a logical circuit, which counts current pulses from both detectors arriving within a time-window of 8 ns as a coincidence and displays the coincidence rate on a computer.

6 Experimental procedure

6.1 Laser safety



The pump laser has a wavelength of 404 nm and operates at a power of 50 mW, which places the laser in class 3B. This means that a direct hit of the laser beam to the eye (including reflections from a mirror or any other shiny surface) is likely to cause permanent damage to the retina. However, as opposed to class 4 lasers, diffuse stray light from matt surfaces is safe. The photon flux from the near-infrared photons is also too weak to cause any harm. Your instructor will give you a laser safety briefing at the start of the experiment. Here is a summary of the most important rules, which you must respect at all times when working with the experiment:

1. Under no circumstances look directly into the pump laser beam path! The pump laser is mostly contained in one half of the setup and enters the other only to pass the crystal before it is absorbed in a beam dump (see Fig. 2).
2. All optical elements (mirrors, lenses, crystal, etc.) are attached to the table. This ensures that the laser beam does not leave its designed path. Do not loosen or shift any element! A tumbling mirror may cause uncontrollable reflections anywhere in the lab. Use only the fine-adjustment screws for alignments, if necessary.
3. Do not wear watches, jewelry or any other shiny surfaces on your hand or wrist! They may just as well cause uncontrollable reflections.
4. There may be other experiments running in the lab at the same time. The scientists working on these are informed of your presence and will make sure that no harmful radiation enters the area where you are working in. In any case, do not enter other zones of the lab on your own and do not stick your head above or into other optical setups (obviously do not touch anything either), as dangerous laser radiation might be present in those spaces.

6.2 Measuring the Visibility

The first task for this experiment is to measure the coincidence rates between the photon pairs for different angles of Polarizers A and B. Figure 4 shows such a measurement, where A was set to a fixed angle and B was rotated. The coincidence rate was recorded for each setting of

B. When A is set to 0° or to 90° the measurement is in the H/V basis and if it is set in 45° or 135° the measurement is in the D/A basis.

1. Set the polarizer A to H and record the coincidence rate for various settings of the (motorized) polarizer B to acquire the correlation curves. Choose the scan range, angle resolution and integration time as you find appropriate.
2. Repeat the process with polarizer A set to the settings D , A and V .
3. Plot the curves and calculate the visibilities.

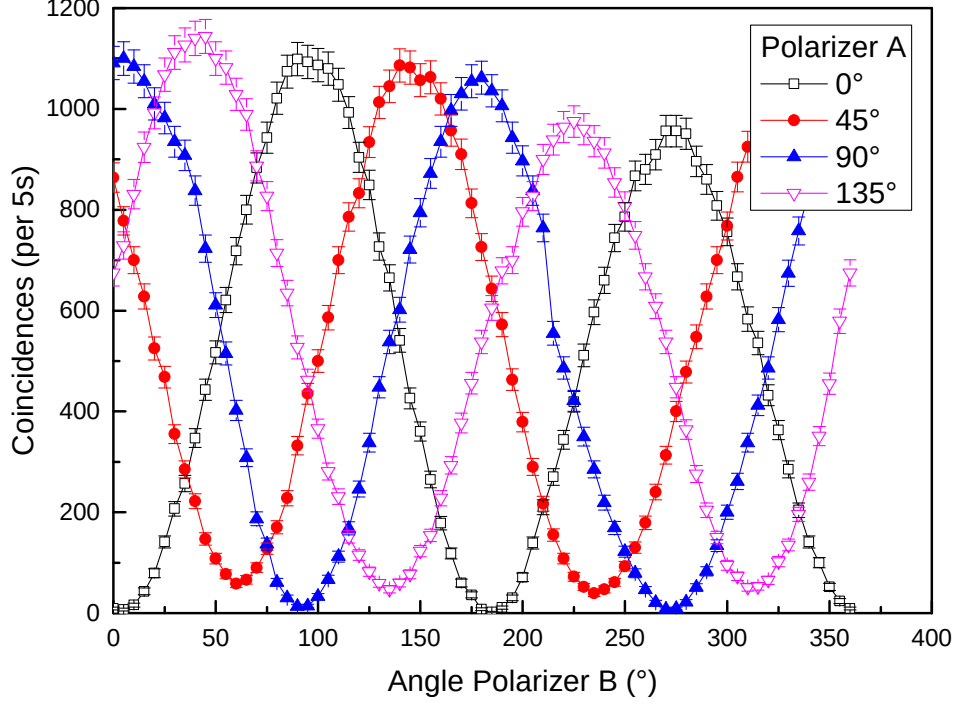


Figure 4: For four different settings of Polarizer A, this figure shows the coincidence count rates as a function of the angle of Polarizer B. The errorbars show the Poissonian standard deviation of the count rates. The lines connecting the symbols are only to guide the eye.

From the data we can then evaluate the visibility, and it is apparent that we will obtain different results for the different curves. The visibility is usually much better for the H/V basis than for the D/A basis. The visibility is defined by:

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

where $I_{\max}$ and $I_{\min}$ correspond to the expected and unexpected coincidence rates for the particular state being produced. In the case of the $|\psi^-\rangle$ state, the expected coincidences are (HV, VH, DA, AD) while the unexpected events (HH, VV, DD, AA) . Therefore, you can rewrite the visibility in the following way:

$$V_{HV} = \frac{N_{HV} + N_{VH} - N_{HH} - N_{VV}}{N_{HV} + N_{VH} + N_{HH} + N_{VV}} \quad (7)$$

and

$$V_{DA} = \frac{N_{DA} + N_{AD} - N_{DD} - N_{AA}}{N_{DA} + N_{AD} + N_{DD} + N_{AA}},$$

respectively, where N_{HV} is the number of coincidences when the polarizer A is in H and the polarizer B is in V , etc.. A visibility of $V_{HV} = V_{DA} = 1$ would be expected for a perfect $|\psi^-\rangle$ state.

6.3 Bell Parameter

The second task is to test the CHSH inequality. To calculate the Bell parameter S you have to measure the coincidence rates for 16 different angle settings with an extended measurement duration to get good counting statistics. Basically you have to fill a table as given in Table 1. With this you can then calculate the expectation values using equation (5) and the Bell parameter with equation (3). Note that the angles in the table refer to our (arbitrary) convention $H \leftrightarrow 0^\circ$, $V \leftrightarrow 90^\circ$, not necessarily to the angles imprinted on the rotation mounts of the polarisers.

Polarizer angle (A/B)	α/β ++	$\alpha/\beta + 90^\circ$ +-	$\alpha + 90^\circ/\beta$ -+	$\alpha + 90^\circ/\beta + 90^\circ$ --
$\alpha = 0^\circ/\beta = 22.5^\circ$				
$\alpha = 0^\circ/\beta = 67.5^\circ$				
$\alpha = 45^\circ/\beta = 22.5^\circ$				
$\alpha = 45^\circ/\beta = 67.5^\circ$				

Table 1: Matrix of required measurements for testing the CHSH inequality.

6.4 Data processing and results

The required data to present in your lab report are

1. Correlation curves, the corresponding visibilities.
2. Value of S for the extended measurement. Check this result for consistency with the visibilities you report earlier.
3. (optional) Variation of the Bell-parameter S with the offset angle ϕ (as defined in Exercise 2).
4. (optional) Variation of the Bell-parameter S with the angle between α and β , but keeping $\alpha' - \alpha = \beta' - \beta = 45^\circ$ constant.

7 Questions and Exercises for your preparation

1. Carry out the exercises 1-3 from the text above.
2. What is the output state behind a linear polarizer oriented at angle α for H - and V -input?

3. Assuming an incoming photon pair in $|\psi^-\rangle$ state, calculate explicitly the coincidence detection probability for linear polarizer angles α and β . Derive Eq. (4) from that.
4. Which probability distribution does the number of detected photons per time interval follow? Which parameters are needed to characterise this distribution? How is it for coincidences?
5. The type-II SPDC source does not produce a perfect state. A good model for the actual state is a mixture of mostly $|\psi^-\rangle$ plus a small $|\psi^+\rangle$ contribution. Write down the density matrix of this state, if the probability of the $|\psi^-\rangle$ state is F . What impact do you expect on the visibilities V_{HV}, V_{DA} and the Bell parameter S ?
6. Estimate the minimum visibility for which the CHSH inequality can still be violated.

Acknowledgements

We thank Benedikt Hilbe for the initial assembly of the optical setup, Katharina Isser, Bernhard Ertel and Ulrich Czopak for various improvements, Zoltan Vörös for the coincidence counter electronics and Manos Amargianitakis for automatising the polariser rotation and data recording of the visibility curves.

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Appendix - Alignment

The experiment is pre-aligned by the instructor, because it would take too much time to do this from scratch in one afternoon. It is strongly advised not to change the alignment unless the visibility and Bell measurements do not yield sensible results (please consult with your instructor, in any case). However, once you have completed those measurements, you may try to optimize the setup further. Please proceed with caution and only make small changes, one at a time. The alignment proceeds as follows (assuming that you already receive some counts on the APDs):

Initially we align for the highest single count rate for both arms (polarisers removed) by adjusting the pointing of the fiber coupler and its preceding mirror as well as by adjusting the focus of the collimating lens. Second one checks with a polarizer, whether the count rate for H and V polarizations are similar. If they differ by much, more alignment is required. This may reduce the overall count rate a bit. As a next step we monitor the coincidence rate and try to maximize it again by adjusting the mirror, the fiber couplers and the coupling lenses on both sides. Once we have a useful number of coincidence counts (a few hundred per second) we check how the $H-V$ and $V-H$ coincidence count rates compare. They should be roughly equal. If not, more alignment is required. In particular we may need to “walk the beam” by using a small transverse movement on one side (with the appropriate angular adjustment) that is “compensated” by a small movement on the other side. If the $H-V$ and the $V-H$ coincidence count rates are equal to within 20%, we can check the $H-H$ and the $V-V$ coincidence count rates. They should be extremely small. If not, alignment of the HWPs might be necessary. After that the polarizers should be set to $D-D$ and/or $A-A$ to optimize the compensation crystal angle (by minimising the coincidences). By adjusting the compensation crystal vertical tilt angle we can set the phase of the state to be $|\psi^-\rangle$ or $|\psi^+\rangle$.

Please do not change the alignment of the pump laser beam path or the orientation of the primary BBO crystal. Realigning those would take a lot of time.